Olympiad Practice Examination
Answer Key and Worked Solutions

Problem 1 — Polynomials

The answers are

$$\boxed{P(x) = c\,x(x-1)(x+1) \qquad (c \in \mathbb{R} \setminus \{0\}).}$$

Substituting $x = 1, -2, 0$ in turn into the given identity gives $P(1) = P(-1) = P(0) = 0$. Hence

$$P(x) = x(x-1)(x+1)Q(x)$$

for some real polynomial Q. Since $P(x+1) = (x+1)x(x+2)Q(x+1)$, substitution and cancellation of the nonzero polynomial $x(x-1)(x+1)(x+2)$ give

$$Q(x+1) = Q(x).$$

Thus $Q(n) = Q(0)$ for every nonnegative integer n. The polynomial $Q(x) - Q(0)$ therefore has infinitely many roots, so Q is constant. Nonconstancy of P forces this constant to be nonzero.

Conversely, for $P(x) = cx(x-1)(x+1)$,

$$(x-1)P(x+1) = c(x-1)x(x+1)(x+2) = (x+2)P(x),$$

so every displayed polynomial is a solution.

Problem 2 — Inequalities

Set $x = 1/a$, $y = 1/b$, $z = 1/c$. Then $xyz = 1$ and

$$\frac{1}{a^3(b+c)} = \frac{x^3yz}{y+z} = \frac{x^2}{y+z}.$$

The other two terms transform similarly. By Cauchy–Schwarz (Engel form),

$$\sum_{\text{cyc}} \frac{x^2}{y+z} \geq \frac{(x+y+z)^2}{2(x+y+z)} = \frac{x+y+z}{2} \geq \frac{3}{2},$$

where the last step is AM–GM together with $xyz = 1$. Equality requires $x = y = z$ and $xyz = 1$, so equality holds exactly at

$$\boxed{a = b = c = 1}.$$

Problem 3 — Functional Equations

The solutions are

$$\boxed{f(x) = a\sin x + b\cos x \qquad (a, b \in \mathbb{R}).}$$

Let $b = f(0)$. Putting $x = 0$ gives

$$f(y) + f(-y) = 2b\cos y. \tag{1}$$

Define $g(x) = f(x) - b\cos x$. Since $b\cos x$ itself satisfies the given equation, g also satisfies it. Moreover, (1) shows that g is odd.

Applying the equation for g first with (x, y) and then with (y, x), and using oddness in the second equation, yields

$$g(x+y) + g(x-y) = 2g(x)\cos y,$$
$$g(x+y) - g(x-y) = 2g(y)\cos x.$$

Adding and dividing by 2, we obtain

$$g(x+y) = g(x)\cos y + g(y)\cos x. \tag{2}$$

Let $a = g(\pi/2)$. Putting $y = \pi/2$ in (2) gives $g(x+\pi/2) = a\cos x$. Replacing x by $x - \pi/2$ therefore gives $g(x) = a\sin x$, and hence $f(x) = a\sin x + b\cos x$.

Finally, the addition formulas for sine and cosine show directly that every function of this form satisfies the original equation.

Problem 4 — Complex Numbers

The numbers ω^k, $0 \le k < n$, are precisely the roots of $z^n - 1$, so

$$z^n - 1 = \prod_{k=0}^{n-1} (z - \omega^k).$$

Taking the logarithmic derivative at $z = 2$ gives

$$\sum_{k=0}^{n-1} \frac{1}{2 - \omega^k} = \frac{n2^{n-1}}{2^n - 1}.$$

Thus the requested value is

$$\boxed{\frac{n2^{n-1}}{2^n - 1}}.$$

Problem 5 — Combinatorics

There are $\binom{2n}{n}$ unrestricted sequences containing n copies of each sign. Call a sequence *bad* if some initial segment contains more $-$ signs than $+$ signs.

In a bad sequence, take the first position at which the running total (with $+$ counted as 1 and $-$ as -1) equals -1. Reflect every symbol *after* that position, interchanging $+$ and $-$. The resulting sequence has $n - 1$ copies of $+$ and $n + 1$ copies of $-$.

Conversely, every sequence with these new counts must reach running total -1 at some first position. Reflecting the suffix after that position reconstructs a bad sequence with n copies of each sign. These operations are inverse bijections. Consequently, the number of bad sequences is $\binom{2n}{n+1}$, and the number of balanced sequences is

$$\binom{2n}{n} - \binom{2n}{n+1} = \binom{2n}{n}\left(1 - \frac{n}{n+1}\right) = \boxed{\frac{1}{n+1}\binom{2n}{n}}.$$

Problem 6 — Pigeonhole Principle

The maximum is $\boxed{n}$.

Every positive integer has a unique representation as $2^j q$ with q odd. For each odd $q \le 2n - 1$, form the chain

$$C_q = \{q, 2q, 4q, \ldots\} \cap \{1, 2, \ldots, 2n\}.$$

These n chains partition the universe, and within any one chain, every two distinct elements are comparable by divisibility. If S had at least $n+1$ elements, the pigeonhole principle would put two elements in the same chain. One would then divide the other, a contradiction. Therefore $|S| \le n$.

This bound is attained by

$$S = \{n+1, n+2, \ldots, 2n\}.$$

Indeed, if $a, b \in S$ and $a < b$, then $2a > 2n \ge b$, so b/a cannot be an integer greater than 1. Thus no element divides a different element, and the maximum is exactly n.

Problem 7 — Graph Theory

A connected graph in which every vertex has even degree has an Euler circuit. Traverse one such circuit and orient each edge in the direction of traversal. Whenever the circuit enters a vertex, it next leaves that vertex, contributing one to its indegree and one to its outdegree. At the starting vertex, the final arrival and initial departure are paired similarly. Hence indegree and outdegree are equal at every vertex.

Problem 8 — Game Theory

The losing positions are precisely the multiples of 3.

Indeed, no power of 2 is divisible by 3, so every move from a multiple of 3 leaves a position not divisible by 3. Conversely, from $n \equiv 1 \pmod 3$, removing 1 stone leaves a multiple of 3; from $n \equiv 2 \pmod 3$, removing 2 stones leaves a multiple of 3. These moves are legal, including at $n = 1, 2$, where they win immediately. Induction on the number of stones proves the classification.

Since $2026 \equiv 1 \pmod 3$, the **first player** has a winning strategy. Remove 1 stone initially. After each opposing move, the position is not a multiple of 3: remove 1 stone if its residue is 1, and 2 stones if its residue is 2. This always returns a multiple of 3 to the opponent. The pile strictly decreases, and eventually the opponent receives 3 stones; after any legal move, the first player removes the remaining 1 or 2 stones and wins.

Problem 9 — Number Theory

The answers are $\boxed{n = 1, 2, 4, 6}$.

Modulo $2^n + n$, we have $2^n \equiv -n$. Therefore

$$8^n + n = (2^n)^3 + n \equiv -n^3 + n = -n(n-1)(n+1).$$

Thus the desired divisibility is equivalent to

$$2^n + n \mid n^3 - n. \tag{1}$$

For $n \ge 10$, we claim that $2^n > n^3$. At $n = 10$, this is $1024 > 1000$. If it holds for some $n \ge 10$, then

$$2^{n+1} > 2n^3 > (n+1)^3,$$

because $(1 + 1/n)^3 \le (11/10)^3 < 2$. This proves the claim by induction. Consequently, for $n \ge 10$,

$$2^n + n > n^3 > n^3 - n > 0,$$

so (1) is impossible. Checking the remaining values gives

n	$2^n + n$	$n^3 - n$	Divisibility?
1	3	0	yes
2	6	6	yes
3	11	24	no
4	20	60	yes
5	37	120	no
6	70	210	yes
7	135	336	no
8	264	504	no
9	521	720	no

which completes the proof.

Problem 10 — Geometry — Analytic

For any point $X = (x, x^2)$ on the parabola,

$$XF = \sqrt{x^2 + \left(x^2 - \frac{1}{4}\right)^2} = x^2 + \frac{1}{4},$$

which is its distance to the line $d : y = -\frac{1}{4}$.

A vertical line through F meets the parabola only once, so the given line is nonvertical. Write it as $y = mx + \frac{1}{4}$. Its intersections with the parabola have abscissae satisfying

$$x^2 - mx - \frac{1}{4} = 0.$$

The discriminant is $m^2 + 1 > 0$, and the product uv of the two roots u, v is $-1/4 < 0$. Hence F, whose abscissa is 0, lies between A and B on the focal chord, giving $AF + BF = AB$.

Let O be the midpoint of AB. The circle with diameter AB has center O and radius $AB/2$. Using the focal-distance calculation above, the distance from O to d is

$$\frac{y_A + y_B}{2} + \frac{1}{4} = \frac{(y_A + \frac{1}{4}) + (y_B + \frac{1}{4})}{2} = \frac{AF + BF}{2} = \frac{AB}{2}.$$

The center's distance to d therefore equals the radius, proving that the circle is tangent to d.

Problem 11 — Geometry — Synthetic

All angles in this argument are directed modulo 180°. Because $PM \perp AB$ and $PD \perp AC$, rotating both AB and AC through 90° gives

$$\angle MPD = \angle(PM, PD) = \angle(AB, AC) = \angle BAC.$$

Using that A, B, C, D are concyclic and that $P \in BD$, $M \in CD$, we get

$$\angle BAC = \angle BDC = \angle(DP, DM) = \angle PDM.$$

Thus $\angle MPD = \angle PDM$, so $MP = MD$.

Similarly, $PM \perp AB$ and $PC \perp PB$ imply

$$\angle MPC = \angle(PM, PC) = \angle(AB, PB) = \angle ABD.$$

Cyclicity and collinearity then give

$$\angle ABD = \angle ACD = \angle(CP, CM) = \angle PCM.$$

Hence $\angle MPC = \angle PCM$, so $MP = MC$. Combining the two length equalities yields $MC = MD$. Since M lies on line CD and is equidistant from C and D, it must be their midpoint. □